4} The Female Reproductive Cycle

(Menstrual cycle)

Goal:

At the end of this lecture every student should acquire complete knowledge about female menstrual cycle.

Objectives:

At the end of this lecture every student should be able to

1. Define the new terms present in the lecture.
2. Define the menstrual cycle.
3. Explain hormonal regulation of the menstrual cycle.
4. Discuss the phases of the menstrual cycle.
5. Explain the female ovarian cycle.
6. Describe the female endometrial cycle.

Outline: -

1. Introduction.
2. Definitions of new terms.
3. Definition of menstrual cycle.
4. Hormones of the menstrual cycle.
5. Female reproductive cycle
   a) Ovarian cycle
   b) Endometrial cycle
6. Facts about the female reproductive cycle (Menstrual Cycle).

Introduction:

An ovarian, or estrus, cycle is a characteristic of all female mammals periodic uterine bleeding that accompanies the ovarian cycle, occurs only in women, and some monkeys. In human females the onset of the menstrual cycle is called **menarche** and represents the start of puberty. This occurs around the age of 12 years, but varies widely between individuals and continues, unless pregnancy and lactation occur until menopause at the age of 45-50 years. There is an average that non- pregnant human female may experience approximately 400 menstrual cycles during her reproductive life, resulting in the maturation and release of about 400 oocytes. The menstrual cycle is one of the physiological functions of the female body and controlled by mechanisms involving hormones released the hypothalamus, the pituitary,

ovaries, adrenal glands and even the thyroid, all have to work together with their different hormonal and functional influences. The average cycle can be divided into two approximately equal parts namely the follicular phase that is the period before ovulation, and the luteal phase, the period after ovulation.

Definitions of new terms: -

1. Oogenesis: It is maturation of female germ cell in the ovary. Or: Formation of gametes (ova) in the female.
2. Gamete: Reproductive cells in the female (an ovum), and in male (spermatozoon).
3. Ovulation: Release of the mature ovum from the ovary.
4. Corpus luteum: Graffian follicle cells remaining after ovulation that produce estrogen and progesterone.
5. Graffian follicles: A small sac within the ovary that contains the maturing ovum.
6. Endometrium: It is the lining of the uterus.

Abbreviations:

- ☞ FSH: Follicle- Stimulating Hormone
- ☞ LH: Luteinizing Hormone
- ☞ GnRH: Gonadotropin-Releasing Hormone

Definition of menstrual cycle:

The menstrual cycle is the monthly series of natural changes in hormone production and the structures of the uterus and ovaries to prepare for the possibility of pregnancy

- ☞ Normal cycle length: **21–35 days** (average: 28 days).

☞ First menstruation is called **menarche**
☞ Cessation or stop of menstruation called **menopause**.

OR: Monthly cyclic pattern of ovulation and menstruation involving:

a) **Ovulation:** discharge of the mature ovum from the ovary.

b) **Menstruation:** Periodic shedding of blood and mucous epithelial cells from the uterus (average blood loss is 50 - 80 ml in a period, duration of bleeding ranges from 2 to 7 days).

Hormonal regulation of the menstrual cycle:

The menstrual cycle is regulated by a complex interplay of hormones, which:

- **Induce ovulation**.
- **Prepare the endometrium:** for potential implantation of a fertilized ovum.

Key Hormones and Their Roles:

1. Gonadotropin-Releasing Hormone (GnRH)

- Produced by the **hypothalamus**.
- Travels via the portal vein to the **anterior pituitary.**
- Stimulates the release of FSH and **LH**.

2. Follicle-Stimulating Hormone (FSH)

- Secreted by the **anterior pituitary**
- Stimulates development of **Graafian follicles** in the ovaries

- Promotes **estrogen secretion** by the follicles

3. Luteinizing Hormone (LH)

- Secreted by the **anterior pituitary.**
- Triggers **ovulation.**
- Stimulates the **corpus luteum** to secrete **progesterone**.

4. Estrogen

- Secreted by **developing follicles**
- Stimulates **endometrial repair and thickening** after menstruation
- Promotes **LH surge** by stimulating the pituitary gland
- Participates in **negative feedback** to regulate FSH and GnRH

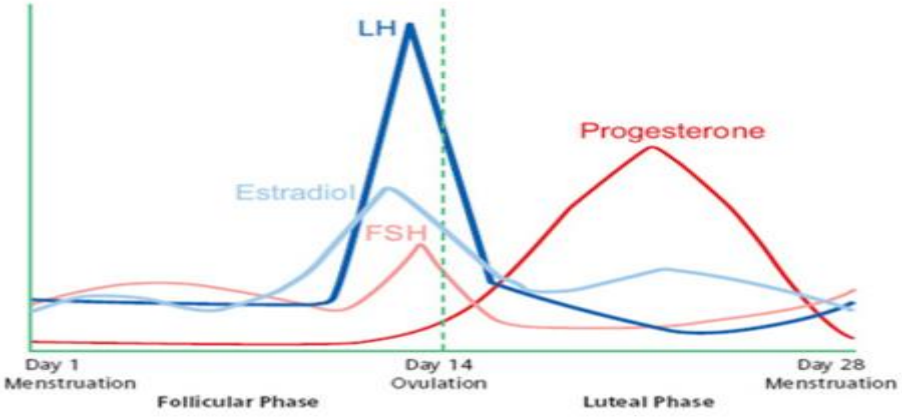
5. Progesterone

- Secreted by the **corpus luteum** after ovulation
- Maintains the **endometrial lining**
- Inhibits **FSH production**
- **Thermogenic**: raises basal body temperature by ~0.5°c at ovulation.

Hormone	Function
GnRH	Stimulates pituitary to release FSH & LH
FSH	Stimulates follicle growth
LH	Triggers ovulation
Estrogen	Builds endometrial lining; feedback to inhibit

	FSH
Progesterone	Maintains endometrium; suppresses further ovulation

Hormone during menstrual cycle



https://pointspecifics.com.au/2020/01/19/what-do-my-hormones-do-throughout-my-cycle/

Components of the female Reproductive Cycle:

The female reproductive cycle is further divided into two cycles that reflect changes in the ovaries and uterine endometrium.

A. Ovarian cycle:

In response to GnRH from the hypothalamus, the anterior pituitary secretes FSH and LH, the FSH and LH stimulate the ovaries to mature an ovum, release it and to secrete other hormones that will prepare the endometrium to implantation of a fertilized ovum.

❖ Phases of the Ovarian Cycle

The **ovarian cycle** consists of three phases and reflects changes in the ovaries in response to hormonal signals:

1. Follicular Phase (Day 1–14):

- Begins on the **first day of menstruation**
- A drop in **estrogen and progesterone** levels triggers the **anterior pituitary** to release FSH and LH.
- FSH stimulates 6–12 follicles to grow; each contains an immature ovum
- Follicles secrete **estrogen**, which:
  - ☞ Stimulates endometrial repair.
  - ☞ Makes follicles more sensitive to FSH.
- One dominant follicle matures and secretes high estrogen levels.
- Rising estrogen suppresses FSH, preventing further follicle maturation.

2. Ovulatory Phase (~Day 14):

- Triggered by a sharp **LH surge** (and a smaller FSH surge)
- LH surge:
  - ☞ Stops estrogen production.
  - ☞ Initiates ovum release (ovulation).
  - ☞ Stimulates formation of the corpus luteum.
- Slight **increase in progesterone** begins during this phase

3. Luteal Phase (Day 15–28):

- The **corpus luteum** forms from the ruptured follicle.
- Secretes **progesterone** (and some estrogen).
- **Progesterone**:
  - ☞ Prepares and maintains the endometrium.

- Inhibits FSH to prevent new follicle development.

- The **functional layer** of the endometrium thickens, peaking around **Day 25**

- If no fertilization occurs:
  - Corpus luteum degenerates.
  - Hormone levels fall.
  - Menstruation begins (Day 28).

B. Endometrial cycle:

The endometrial cycle describes the changes that occur in the lining of the uterus (endometrium) under the influence of estrogen and progesterone.

❖ Phases of the Endometrial (Uterine) Cycle

It consists of **three main phases**:

1. Menstrual Phase (Days 1–5)

- The **functional layer** of the endometrium is **shed** due to falling levels of estrogen and progesterone (after corpus luteum degeneration).
- This is the visible **menstrual bleeding** (which combined with small pieces of endometrial tissue).
- Decidua functional is 2/3 superficial, that proliferates and sheds every cycle.
- Decidua basalis is the deepest region and the source of endometrial regeneration after each menses.

Hormonal Context:

- Low estrogen and progesterone
- FSH starts to rise to initiate new follicle development.

2. Proliferative Phase (Days 6–14)

- ☞ **Estrogen**, secreted by developing ovarian follicles, **stimulates regeneration (repair) and thickening** of the endometrial lining (2-3mm thicker).
- ☞ Endometrial glands and blood vessels proliferate (increase in number and length) and the spiral arteries elongate.

Hormonal Context:

- Rising estrogen levels
- Prepare uterus for possible implantation

🕒 **Timing:** Ends at **ovulation** (Day 14 in a 28-day cycle)

3. Secretory Phase (Days 15–28)

- ☞ After ovulation, the corpus luteum secretes progesterone, which causes the endometrium to become more glandular, edematous, and vascular.
- ☞ Spiral arteries grow into superficial layer and become more coiled
- ☞ Large venous network develops.
- ☞ Glands secrete nutrients to support early embryo if fertilization occurs.

Hormonal Context:

- High progesterone (and moderate estrogen).
- If no fertilization → hormone levels fall → menstruation begins again.

If fertilization occurs:

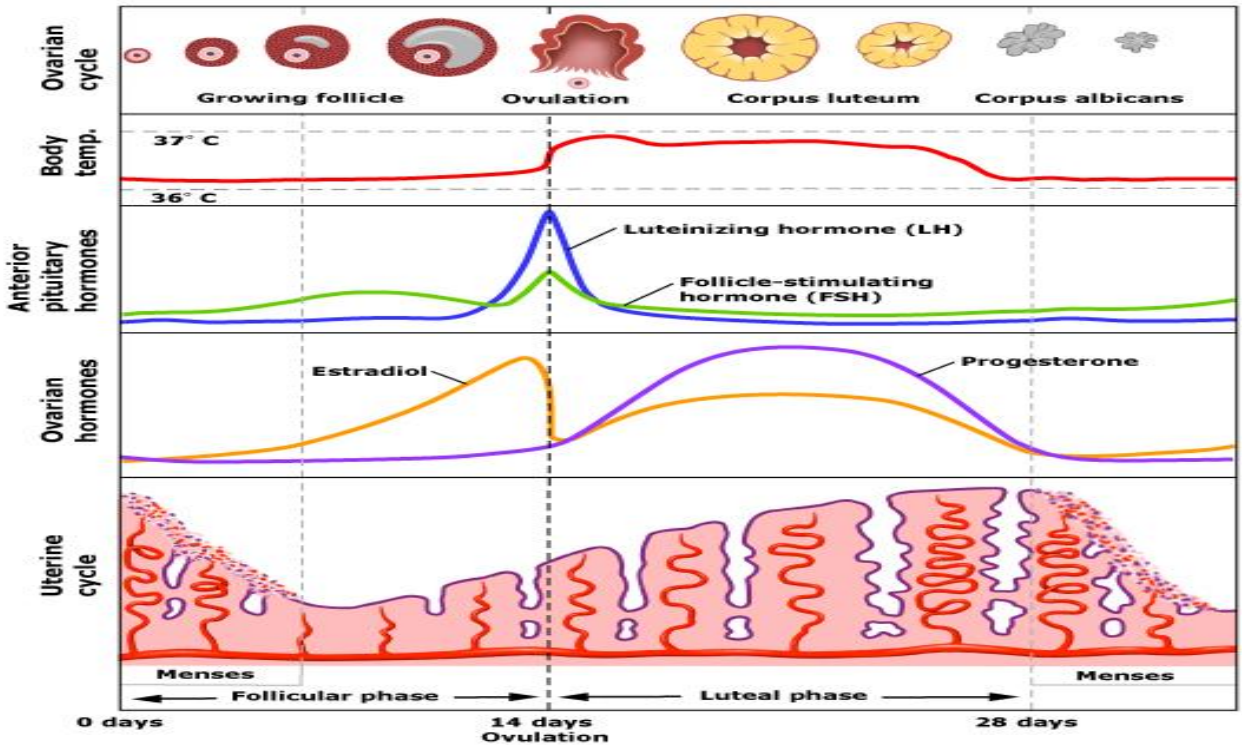
- Fertilized ovum implants in endometrium about 6th day of this phase.
- HCG hormone secreted by fertilized ovum keeps the corpus luteum secreting estrogen and progesterone.

If fertilization doesn't occur:

- No HCG- corpus luteum degenerates- drop in estrogen and progesterone levels- ischemic endometrium- menstruation occurs.

Cycle Phase	Uterine Changes
Menstrual Phase	Shedding of endometrial lining (Days 1–5)
Proliferative Phase	Estrogen promotes endometrial thickening (Days 6–14)
Secretory Phase	Progesterone maintains thick endometrium (Days 15–28)
Ischemic (Premenstrual) Phase	Drop in hormones → endometrial breakdown → menses (2 days before menses)

Menstrual cycle graph



https://www.timeofcare.com/the-menstrual-cycle/

Nursing Considerations/ role:

- **Assessment:** Menstrual history, cycle length, flow, pain, associated symptoms.
- **Education:**
  - Normal vs abnormal patterns.
  - Importance of hygiene during menstruation.
  - Recognizing symptoms that need referral.
  - Cycle tracking methods (apps, calendars).
- **Supportive care:**
  - Pain management (NSAIDs, heat).
  - Lifestyle advice (diet, exercise, stress reduction).